FARM-TO-MARKET HARYANA

Smart Daily Vegetable Trade & Mandi Intelligence Platform

Complete Problem → Solution → AI/Decision Engine → Arhtiya Data → Transport → Buyer → Sale Blueprint

Primary use-case

Daily fresh-vegetable trading: a farmer, trader or aggregator has produce ready today and must decide which mandi is comparatively suitable, how much transport will cost, which market-side participant is available, and what the likely net outcome will be.

Core answer the platform provides:

“Aaj meri sabzi kis mandi mein le jaana comparatively better rahega, kyun, kitna transport lagega, aur expected net outcome kya ho sakta hai?”

This document merges the original Farm-to-Market Haryana blueprint with the refined daily fresh-vegetable mandi model and the registered arhtiya daily-rate contribution module.

1. Executive Concept

The platform is not another mandi-price portal. Its main purpose is to turn scattered daily market information into a practical selling decision and then help coordinate the chosen route.

The first and strongest prototype scenario is fresh vegetables traded every day/night. Example: cauliflower is loaded at night and the trader is considering Kurukshetra, Kaithal, Karnal, Sonipat or Panipat. A higher displayed rate is not automatically the best choice because distance, transport fare, arrival pressure, demand, handling and spoilage can change the actual economics.

The system therefore combines verified daily mandi inputs, transport information, route distance/time, demand/arrival signals, produce quality and urgency. It returns a ranked set of options with an explainable score and an estimated net outcome.

Important: AI does not invent the mandi rate. Registered market participants such as verified arhtiyas/mandi-side users can enter daily rates and market conditions. The recommendation engine uses those submitted inputs along with other available data.

2. The Actual Problem

The original blueprint identifies several economic problems; for the refined version, these are applied specifically to the daily fresh-vegetable trading ecosystem.

Problem	What happens in real trading
Wrong mandi decision	Trader knows one rate but may not know whether another mandi currently offers a better overall opportunity.
Rate is not equal to profit	A distant mandi with a higher rate may become worse after transport, handling and other known costs.
Daily rate changes	Fresh vegetables are highly time-sensitive. A rate seen earlier may not represent the current market situation.
High arrival / weak demand	A mandi can have a good-looking rate but heavy arrivals may create selling pressure and higher risk of a weak r
Transport uncertainty	Finding the right vehicle at the required time, capacity and fare can be difficult.
Arhtiya coordination	Market-side participants are important to daily selling, but their availability and market information may be fragr
Spoilage / urgency	Vegetables cannot always wait. Delay can reduce quality and sale value.
Small quantity / bargaining	Individual lots may have weaker bargaining power; aggregation can create a stronger collective lot.

The platform should therefore solve decision + coordination, not simply display a number.

3. Who Uses the Platform?

Role	Main responsibility / benefit
Farmer	Adds produce, quantity, quality, urgency and location; compares mandis/routes; requests transport; selects sel
Daily Vegetable Trader / Aggregator	Can compare markets before dispatching produce and coordinate transport and market-side selling.
Registered Arhtiya / Mandi-side User	Provides daily mandi rate and market-condition inputs; can participate in market coordination.
Transporter	Lists vehicle, capacity, availability, route preference and estimated fare; accepts trip requests.
Buyer / Processor	Posts requirements, quantity and quality needs; receives suitable produce opportunities.
FPO / Collective Group	Combines nearby lots to improve quantity, transport efficiency and buyer access.
Admin	Verifies roles, monitors data quality and views anonymised market/transport intelligence.

4. Registered Arhtiya Daily Mandi-Rate Module

This is a major addition for the refined project. The system should not claim that its AI independently knows the live mandi rate. Instead, verified registered arhtiyas/mandi-side users can contribute daily market information for the mandi they are associated with.

Role positioning: Arhtiya is not treated as an “evil middleman”. In the prototype, the arhtiya is a verified market participant and a source of market-side information, while also being part of the selling ecosystem.

Field	Example / purpose
Mandi	Karnal Mandi
Vegetable	Cauliflower
Minimum rate	₹2,000/quintal
Modal / indicative rate	₹2,300/quintal
Maximum rate	₹2,500/quintal
Arrival quantity	Example: 180 quintal
Demand	Low / Medium / High
Market condition	Weak / Normal / Strong
Trend	Rising / Stable / Falling
Remarks	Optional short market-side note
Timestamp	Updated at 10:42 PM

Multiple arhtiyas: If several registered users submit rates for the same mandi and vegetable, the system can show an indicative consensus/range. If submissions differ greatly, the platform should reduce confidence and flag the data for review.

Freshness: Every rate must show its update time. Example: “Updated 12 minutes ago.” Old data should not be presented as live.

Verification: Only verified accounts assigned to the relevant market should be allowed to publish market-rate inputs. Unusual entries can be flagged instead of being silently accepted.

5. Complete System Workflow

End-to-end flow:

Farmer / Trader → Add Vegetable Lot → Select Target Time → Load Daily Mandi Data → Add Transport Options → AI/Decision Engine → Compare Mandis → Expected Net Outcome + Market Risk → Top 3 Recommendations → Select Mandi → Find/Book Vehicle → Coordinate Market-side Participant → Deliver/Sell → Record Outcome → Improve Intelligence

1. User registers and selects role: Farmer, Trader/Aggregator, Arhtiya, Transporter, Buyer/Processor, FPO or Admin.
2. User adds vegetable, quantity, location, quality/grade, harvest/loading time and urgency.
3. System collects available daily mandi data from registered/verified arhtiya entries and other permitted sources.
4. System collects transport options: vehicle capacity, estimated fare, availability and destination.
5. Decision engine calculates comparable economics for candidate mandis.
6. System considers arrival, demand, freshness of rate, distance/time, quality fit and spoilage/urgency.
7. Top 3 mandis are shown with reasons, expected net range and risk/confidence.

8 User chooses a mandi; platform helps find/request a suitable vehicle.

9 The transaction timeline records pickup, destination, delivery and sale outcome.

10 Anonymised outcomes can later support market intelligence and model improvement.

6. Daily Mandi Intelligence Dashboard

The dashboard should be designed for a quick daily decision rather than a complicated analytics screen.

Mandi	Rate	Arrival	Demand	Trend	Data freshness	Risk
Karnal	₹2,300	High	Medium	Falling	8 min	Medium-High
Kaithal	₹2,380	Medium	High	Stable	12 min	Low-Medium
Kurukshetra	₹2,450	Medium	High	Rising	15 min	Low
Panipat	₹2,520	Low	High	Rising	9 min	Low
Sonipat	₹2,500	Low	Medium	Stable	21 min	Medium

Important: These figures are illustrative prototype data only. They must not be presented as real-time government rates unless the system actually receives authorised/live data.

The user should be able to filter by vegetable, quantity, maximum travel time, preferred district/mandi and urgency.

7. AI / Decision Engine

For the competition MVP, the best approach is a transparent scoring/recommendation engine first. A production-grade ML model is optional later.

AI input layer: daily registered arhtiya rates + arrival + demand + trend + rate freshness + distance + estimated transport fare + travel time + quantity + quality + urgency + buyer/processor opportunities + storage option where relevant.

AI output: Top 3 mandi/route options, expected net range, market risk, confidence/data freshness and a short explanation of why each option ranked where it did.

Signal	Effect on recommendation
Higher current rate	Positive, if the data is fresh and quality/quantity are suitable.
High arrival	Can increase selling-pressure risk; should not automatically mean rejection.
High demand	Positive signal for sale suitability.
Falling trend	Risk increases if the produce is urgent/time-sensitive.
Long distance	Reduces net outcome through transport and time/spoilage exposure.
High transport fare	Reduces net outcome.
Fresh rate data	Higher confidence than stale data.
Quality mismatch	Reduces route suitability.
Urgent sale	Favors faster routes and can penalise distant/uncertain options.

8. Mandi Risk & Confidence

The system should never say “this mandi will definitely give low price.” Instead it should communicate comparative risk based on available submitted data.

Example: “Panipat — Low estimated selling risk” means the current submitted rate, demand, arrival, freshness and route economics look comparatively favourable. It is not a guarantee.

Confidence: High when multiple recent verified entries agree; Medium when data is limited; Low when data is stale or highly inconsistent.

Stale-data warning: “Rate updated 7 hours ago — verify before dispatch.”

9. Core Economic Formula

Expected Net Realisable Value (ENRV)

Expected Net Realisable Value = Expected Selling Value − Transport − Storage − Packaging/Handling − Other Known Costs

For the fresh-vegetable version, the system can additionally display an estimated spoilage/time risk as a separate risk signal. It should not silently invent a monetary loss unless a transparent, configurable assumption is being used.

Illustrative example: 100 quintal × ₹2,500 = ₹2,50,000 gross. If transport = ₹6,000 and handling/other known costs = ₹2,000, expected net before any optional risk adjustment = ₹2,42,000.

The key principle is: compare net outcome, not just mandi rate.

10. Routes the Platform Can Compare

Route	Use case
Mandi A	Nearby mandi with current submitted rate and market condition.
Mandi B / C	Alternative mandis with different rates, demand, arrivals and travel costs.
Direct buyer	Buyer purchases directly based on crop, quantity and quality.
Processor	Suitable processor may offer a direct route when quality/quantity fit.
Store and sell later	Applicable only when storage is available and expected value justifies cost/risk.
FPO / collective lot	Combine nearby lots for better buyer access or shared transport.
Wastage rescue / alternate channel	For unsold or rejected produce, identify suitable permitted processing/value-addition alternatives.

11. Transport & Route Coordination

A strong feature is Farm-Gate / Trader-Gate Vehicle Matching. The platform should help the user find a suitable vehicle instead of forcing them to search manually through multiple calls.

Vehicle data	What to store
Owner	Name/phone, verification status
Vehicle	Tractor, pickup, mini-truck, truck, etc.
Capacity	Quintal/ton capacity
Availability	Available date/time
Location	Current/starting area
Destination preference	Preferred routes/mandis
Fare	Estimated trip fare or configurable rate
Reliability	Prototype rating / completed trips

Shared transport: nearby traders/farmers going toward the same mandi can be grouped where feasible, reducing per-lot transport cost.

12. Buyer, Processor, Storage & FPO Modules

- Direct Buyer Matching: match vegetable + quantity + quality + location with buyer requirements and offer.
- Processor Matching: identify suitable direct purchase opportunities when a processor can provide a better practical route.
- Storage Decision: compare sell-now versus store-and-sell-later using storage cost, expected price range and uncertainty.
- FPO / Collective Lot: combine nearby quantities to approach larger buyers and use shared transport.
- Wastage Rescue: for unsold/rejected produce, surface suitable processing/value-addition or permitted alternative channels.

13. Proposed App Screens

ID	Screen	Purpose
S1	Landing / Role Selection	Farmer | Trader | Arhtiya | Transporter | Buyer/Processor | FPO | Admin
S2	Farmer/Trader Home	My lots, sell today, mandi intelligence, vehicles, buyers, transactions
S3	Add Vegetable Lot	Vegetable, quantity, location, quality, loading time, urgency
S4	Daily Mandi Dashboard	Mandi rates, arrivals, demand, trend, freshness, confidence
S5	Arhtiya Dashboard	Assigned mandi, daily rate entry, history, market condition, pending updates
S6	Best Mandi / Route	Top 3 recommendations with expected net and risk
S7	Route Details	Rate − transport − known costs + distance/time + explanation
S8	Find Vehicle	Nearby vehicles, capacity, fare, availability, verification
S9	Vehicle Request	Pickup, destination, time, quantity, accept/reject
S10	Buyer Marketplace	Buyer requirements and matching opportunities
S11	Storage	Nearby storage, capacity, cost, sell-now vs store
S12	Collective Lot	Combine farmers/traders, target buyer, shared transport
S13	Transaction Tracking	Requested → Accepted → Pickup → Delivered → Sold
S14	History	Past rates/routes, estimated vs realised outcomes
S15	Admin Intelligence	Data quality, district market patterns, transport gaps, surplus opportunities

14. Database Design

Table	Important fields
users	id, role, name, phone, village, district, verification_status
farms	id, farmer_id, location, area, soil_type
produce_lots	id, owner_id, vegetable, quantity, quality, loading_time, urgency
markets	id, name, location, district, active_status
market_rate_entries	id, market_id, arhtiya_id, vegetable, min_rate, modal_rate, max_rate, arrival_qty, demand, condition, trend, remarks, update

Table	Important fields
market_rate_consensus	market_id, vegetable, indicative_rate, min/max range, confidence, last_updated
arhtiya_profiles	id, user_id, market_id, verification_status, active_status
buyers	id, business_name, vegetables, required_quantity, quality, offer_price, location, verified
vehicles	id, owner_id, type, capacity, current_location, destination_preferences, rate, availability, verified, rating
transport_requests	id, requester_id, vehicle_id, pickup, destination, quantity, status
storage	id, name, type, location, capacity, available_capacity, rate
fpo_lots	id, vegetable, total_quantity, members, target_buyer, status
routes	id, lot_id, destination, route_type, gross_value, transport_cost, known_costs, net_value, score, risk, confidence
transactions	id, owner_id, buyer_id, vehicle_id, quantity, final_price, status, timestamps

15. Recommended Technology Stack

Layer	Recommended technology
Frontend	React + Vite + Tailwind/CSS + Lucide; responsive, farmer-first UI
Backend	Node.js + Express for fast prototype, or Spring Boot if the team already uses Java
Database	MySQL
AI / Decision Engine	Python + FastAPI; transparent scoring/rules first, ML later
Maps / Distance	Mapping API if available; prototype can use stored coordinates/sample distance calculations
Authentication	OTP/email mock for MVP; secure role-based authentication for production
Deployment	Vercel for frontend; Render/Railway/AWS or equivalent for backend; hosted MySQL

16. Suggested Mandi Recommendation Logic

Start with a transparent weighted score. Do not hide the reasoning inside an unexplained ML model.

Component	Illustrative weight	Why
Net economic outcome	40%	Main objective: what remains after known costs.
Rate freshness / confidence	15%	Recent and consistent market data is more trustworthy.
Demand + arrival condition	15%	Market pressure affects practical selling suitability.
Transport cost + time	15%	Distance and fare affect net outcome and freshness.
Quality / buyer fit	10%	A route is better when the produce matches the market requirement.
Urgency / spoilage exposure	5%	Time-sensitive vegetables need faster, lower-risk routes.

Weights should remain configurable. The prototype can show the component scores so a judge can understand why the recommendation changed.

17. Strong Hackathon Demo Scenario

1. A vegetable trader/farmer from Haryana enters: Cauliflower, 100 quintal, Grade A, loaded tonight, sale required tomorrow.
2. The system checks the available daily entries for Kurukshetra, Kaithal, Karnal, Sonipat and Panipat.

3. Registered arhtiyas have submitted current rate, arrival, demand, condition and timestamp for their respective markets.
4. The platform compares the rate with transport fare, route time, distance and produce urgency.
5. Example result: Panipat has the highest indicative rate, but Kaithal may have the best overall net outcome because transport is cheaper and demand is strong.
6. The screen shows Top 3: 1) Kaithal — Best Overall, 2) Panipat — Highest Rate but higher travel cost, 3) Kurukshetra — Backup.
7. The user opens Find Vehicle and sees verified available vehicles with capacity and estimated fare.
8. The user sends a vehicle request and the transporter accepts.
9. The app creates one timeline: Vehicle Confirmed → Pickup → Destination Mandi → Sale Recorded.
10. The final sale outcome can later be compared with the earlier estimate for improving the decision engine.

18. Difference from Existing Government / Market Systems

The platform should be positioned as a complementary decision and coordination layer, not as a replacement for government market infrastructure.

System / mechanism	Primary role	Our opportunity
e-NAM	Electronic trading, price discovery and related market infrastructure.	Use available authorised market information as an input; focus on the
Haryana procurement / e-Kharid ecosystem	Government procurement workflow and related processes.	Focus on daily open-market vegetable trading decisions and coordina
Farm-to-Market Haryana	Compare daily market economics and coordinate trade decisions.	Take the last decision: verified arhtiya inputs + transport + demand

Do not say: “Government has no mandi data.”
Say: “Existing systems provide valuable market infrastructure/data; our platform adds a farmer/trader-first decision and coordination layer for the daily fresh-vegetable journey.”

19. What NOT to Build in the First Prototype

- Do not build a full national mandi replacement.
- Do not claim live government data unless the actual API/source is available and authorised.
- Do not present arhtiya-submitted prototype data as government-verified data unless formally verified.
- Do not promise guaranteed future prices.
- Do not build real payments, KYC or legal verification unless the competition requires it.
- Do not start with satellite/IoT/computer vision. These can be future layers.
- Do not overload the farmer/trader UI with dashboards; the primary journey should take only a few clear steps.
- Do not hard-code commission percentages; make charges configurable and based on verified applicable rules/data.

20. Build Order — Step by Step

Phase	Build
1. Foundation	Repository, frontend/backend folders, MySQL schema, roles, seed/sample Haryana mandi data.

Phase	Build
2. User roles	Farmer/trader, arhtiya, transporter, buyer, FPO and admin login flows.
3. Daily mandi module	Market list, rate entries, arhtiya dashboard, timestamps, multiple-entry aggregation and confidence.
4. Mandi intelligence	Rate comparison, arrival/demand/trend cards, freshness and risk.
5. Route engine	ENRV calculator + transparent Top 3 mandi recommendation.
6. Transport	Vehicle profiles, availability, matching, request and accept/reject flow.
7. Buyer/processor	Requirements, matching and direct-sale opportunity.
8. Storage/FPO	Sell-now vs store and collective lot flows.
9. Transaction	Pickup → destination → sale → outcome recording.
10. AI upgrade	Only after deterministic logic works: historical trend/demand model and calibration.
11. Demo polish	Hindi-first labels, icons, realistic sample data, loading/error states and one complete scenario.

21. AI/ML Roadmap

Stage 1 — Rules/Scoring: use submitted daily rates and transparent weighted scoring. This is easiest to demonstrate and audit.

Stage 2 — Historical analytics: calculate market-wise vegetable trends, volatility and demand patterns from stored outcomes.

Stage 3 — ML prediction: estimate a short-term price range or demand signal using historical features. Always show uncertainty.

Stage 4 — Learning from outcomes: compare recommended route vs actual sale, transport cost and time-to-sell; use these outcomes to improve calibration.

Stage 5 — Advanced matching: improve transporter/buyer matching based on successful past transactions, availability and route patterns.

22. MVP Success Metrics

- Average estimated net-value improvement versus the user's default/local route.
- Average transport cost saved through matching/shared trips.
- Time from “produce ready” to confirmed vehicle/market route.
- Percentage of vegetable lots successfully matched to a suitable market/buyer.
- Rate-data freshness and percentage of active verified market contributors.
- Difference between estimated and realised sale outcome.
- Quantity of produce diverted from unsold/waste to suitable processing or alternative channels.
- Number of completed route selections and transactions in the prototype simulation.

23. Judge Questions — Ready Answers

Question	Answer
“Isn't this e-NAM?”	No. We complement market infrastructure. Our focus is the daily decision layer: which mandi/route is comparative
“Where does your rate data come from?”	For the prototype, from clearly labelled sample/seed data and registered/verified arhtiya market entries. Productio
“Why would arhtiyas join?”	They become verified market participants, can publish daily market information and receive relevant market-side

Question	Answer
“What if the arhtiya enters the wrong rate?”	Use role verification, timestamps, multiple-entry consensus, anomaly checks, confidence scoring and audit history
“What if AI is wrong?”	The system shows an estimated range, confidence and alternatives; it never guarantees a price. The farmer/trade
“Why will transporters join?”	They receive local trip requests and can improve vehicle utilisation.
“What does government gain?”	Anonymised aggregate data can reveal market pressure, transport gaps, surplus patterns, storage needs and pro

24. Final System Architecture

A simple architecture for the MVP:

USER APP / WEB → BACKEND API → MYSQL DATABASE
↓
DAILY MARKET DATA LAYER
Arhtiya Entries + Seed/Authorised Data
↓
AI / DECISION ENGINE
Rate + Arrival + Demand + Freshness + Transport + Distance + Time + Quality + Urgency
↓
TOP 3 MANDIS / ROUTES + NET VALUE + RISK
↓
VEHICLE → MARKET-SIDE COORDINATION → SALE
↓
OUTCOME / ANALYTICS / LEARNING

25. Final Product Statement

Farm-to-Market Haryana is a daily fresh-vegetable trade intelligence and coordination platform that converts market-side information into an actionable selling decision.

Given a vegetable, quantity, location, loading time and urgency, the platform compares available mandis using registered daily arhtiya inputs, rate freshness, arrival, demand, transport cost, route time, quality fit and other known costs. It then recommends the Top 3 comparatively suitable routes, explains the reason, helps find a vehicle and records the final transaction outcome.

> The central innovation is not “showing a mandi price.” The innovation is helping the trader/farmer decide where to go today and how to execute that decision with better economic visibility.

The platform remains transparent: rates are attributed to their data source, AI recommendations show confidence and alternatives, and no future price or profit is guaranteed.

Source basis: This document incorporates the provided “Farm-to-Market Haryana — Prototype Blueprint” and the refined daily fresh-vegetable/arhtiya workflow specified in the project discussion. The original blueprint explicitly recommends complementing existing government/market systems rather than duplicating them.

Government/API availability, licensing, current mandi rules and applicable market/commission practices should be verified before production deployment.